

Takshak Singhania

takshaksinghania1@gmail.com | (+91) 9351595646 | [LinkedIn](#) | [GitHub](#) | [CodeChef](#)

SUMMARY

Software Engineering student at IIIT Bhopal with experience building web applications and full-stack projects using **JavaScript, TypeScript, React, Node.js**, and **REST APIs**. Experienced in **collaborative development, algorithmic problem solving, testing**, and **CI/CD** through multi-person projects and development teams. Strong foundation in **data structures, algorithms, operating systems**, and **object-oriented programming**.

EDUCATION

Indian Institute of Information Technology, Bhopal

Sept 2023 – Jun 2027

B.Tech. in Electronics and Communication Engineering

Relevant Coursework: **Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Database Management**

EXPERIENCE

Axios Development Club, IIIT Bhopal

Sep 2025 – Present

Web Development Lead | Part-Time

- Managed **6-member team** using Agile sprints; improved code quality through GitHub-based peer reviews and CI checks.
- Streamlined deployment workflows by automating build processes with **GitHub Actions** and Git version control.

PROJECTS

PayFlow | TypeScript, React, Node.js, PostgreSQL, Redis, BullMQ, Docker

[GitHub](#)

- Architected a **6-state payment lifecycle engine** with PostgreSQL transactions and row-level locking (**SELECT FOR UPDATE**), preventing state inconsistencies across concurrent capture and refund requests.
- Enforced **distributed idempotency** via SHA-256 request hashing and PostgreSQL unique constraints, safely deduplicating concurrent submissions and resolving **error 23505 race conditions** without 500s.
- Engineered an asynchronous webhook delivery pipeline with **Redis and BullMQ** featuring **5-tier exponential backoff retries**, constant-time HMAC-SHA256 signing, and **58 automated unit and integration tests**.

Routewise | React, Node.js, Express, JavaScript

[GitHub](#)

- Engineered an in-memory spatial graph engine using OpenStreetMap vector data across **5 global downtowns**, modeling **14,000+ street nodes** and calculating great-circle distances via the **Haversine formula**.
- Accelerated **Dijkstra and A* pathfinding** by implementing a custom binary MinHeap priority queue, achieving $O(\log V)$ frontier node extraction and animated step-by-step vertex exploration.
- Optimized multi-stop delivery tours by combining a **Nearest Neighbor** heuristic with **2-Opt local search**, iteratively eliminating crossing road segments across **up to 10 waypoints**.
- Constructed **dynamic algorithm benchmarks** comparing heuristic runtimes against exact brute-force ($O(N!)$) solvers, validated by **10 automated unit and integration tests**.

Sentosa QR Ordering System | React, TypeScript, Node.js, Express, PostgreSQL, Prisma, Socket.IO, Razorpay

[GitHub](#)

- Architected a **mobile-first QR ordering platform** with React and Node.js, generating **16-character cryptographic tokens** to secure table sessions and prevent unauthorized cross-table order tampering.
- Developed an authoritative **server-side price integrity engine** with PostgreSQL and Prisma ORM, recalculating subtotals, multi-group modifiers, and **5% GST** directly to eliminate client price manipulation.
- Streamlined kitchen operations by implementing full-duplex WebSocket dispatch via **Socket.IO** with room scoping, live audio alerts, and **2-role RBAC** verified across **37 permission endpoints**.
- Integrated **Razorpay payments** with HMAC-SHA256 signature verification and idempotent webhooks, establishing **63 automated tests** across auth, security, and order lifecycle flows.

TECHNICAL SKILLS

- Languages:** JavaScript (ES6+), TypeScript, SQL, C/C++, Java, Go
- Frontend:** React.js, HTML5, CSS3, Tailwind CSS, Material UI, Redux Toolkit, Responsive Design, DOM Manipulation
- Backend:** Node.js, Express.js, REST APIs, BullMQ, Asynchronous Queues, Full Stack Development, API Security
- Database & Caching:** PostgreSQL, Redis, MongoDB, Relational Schema Design, Query Optimization, Row-Level Locking
- Testing:** Jest, Supertest, Unit Testing, Integration Testing, Postman
- Tools & DevOps:** Docker, Docker Compose, Git, GitHub, Linux, CI/CD, GitHub Actions, Agile/Scrum, VS Code

ACHIEVEMENTS

- Successfully solved **300+ questions** on LeetCode, mastering advanced data structures and algorithmic concepts.
- Achieved a **3-star rating** on CodeChef, demonstrating strong problem-solving abilities and competitive programming skills.